

Metal Prod Service — Comprehensive Website Architecture, Content & Engineering Specification

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Project: Official Corporate & Precision Engineering Web Platform for **Metal Prod Service**
Classification: Engineering Specification, Content Architecture & Quality Assurance Blueprint

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1. Executive Summary & Brand Philosophy

Metal Prod Service is an established, high-precision manufacturing machine shop and precision engineering provider. Rooted in traditional family values, seriousness, reliability, and human craftsmanship, the company couples generational engineering mastery with advanced computerized numerical control (CNC) machining centers and specialized metallurgical heat treatment.

Core Corporate Tenets:

- 1. **Family Tradition & Personal Accountability:** Founded as a family-run enterprise, every partner, client engineer, and procurement manager works directly with experienced professionals dedicated to technical precision.

2. **Dual-Material Mastery:** Complete mastery over standard/exotic alloys (structural steels, tool steels, stainless steels, brass, aluminum, copper) and industrial engineering plastics (POM/Delrin, PTFE/Teflon, PVC, PA6/Nylon).
 3. **End-to-End Cycle:** In-house machining, thermal processing (quenched/tempered via professional industrial furnaces), surface deburring, threading, and coordinate metrology inspection.
 4. **Market-Leading Price/Quality Ratio:** Lean overhead, agile setup protocols, and zero-defect quality philosophy delivering top-tier components at competitive market costs.
-

2. Brand Identity & Logo System

2.1 Design Philosophy

The **Metal Prod Service** logo fuses the mechanical precision of CNC manufacturing with warm, approachable corporate values:

- **Symbolism:** An integrated monogram combining the **M** of *Metal* and **P** of *Prod*, stylized as a rotating 4-flute CNC end-mill cutter intersecting a circular high-tolerance micrometric gauge.
- **Color Palette:**
 - **Anthracite Carbide** (#1E293B): Reflects industrial strength, steel rigidity, and machine foundations.
 - **Precision Cyan / Electric Blue** (#0284C7): Signifies digital CNC controls, sub-micron precision, and modern technology.
 - **Heat-Treat Amber / Flame Gold** (#F59E0B): Reflects industrial hardening furnaces, metallurgical passion, family warmth, and dedication.
 - **Pure Clean Room White** (#FFFFFF): Reflects metrology cleanliness and transparency.

2.2 Vector Logo Code (Embedded Scalable SVG)

```
<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 540 120" width="100%" height="120" aria-label="Metal Prod Service Logo">
  <defs>
    <linearGradient id="metalGrad" x1="0%" y1="0%" x2="100%" y2="100%">
      <stop offset="0%" stop-color="#0284C7" />
      <stop offset="100%" stop-color="#0369A1" />
    </linearGradient>
    <linearGradient id="flameGrad" x1="0%" y1="100%" x2="100%" y2="0%">
      <stop offset="0%" stop-color="#D97706" />
      <stop offset="100%" stop-color="#F59E0B" />
    </linearGradient>
  </defs>

  <!-- Icon Graphic: Integrated Endmill & Calibration Gauge -->
  <g transform="translate(15, 10)">
    <!-- Outer Precision Ring (Micrometer Caliper Ring) -->
    <circle cx="50" cy="50" r="46" fill="none" stroke="#334155" stroke-width="2.5" stroke-dasharray="4 3" />

    <!-- Heavy Solid Base Ring Segment -->
    <path d="M 20,50 A 30,30 0 1,1 80,50" fill="none" stroke="url(#metalGrad)" stroke-width="6" stroke-linecap="round" />

    <!-- Stylized Geometric Flutes / 'M' Cutting Teeth -->
    <path d="M 26 65 L 36 32 L 50 52 L 64 32 L 74 65" fill="none" stroke="#0F172A" stroke-width="5" stroke-linecap="round" stroke-linejoin="round" />

    <!-- Thermal Core Spark / Hardening Flame (Family Warmth & Furnace Heat) -->
    <circle cx="50" cy="38" r="5.5" fill="url(#flameGrad)" />

    <!-- Center Coordinate Crosshairs -->
    <line x1="50" y1="10" x2="50" y2="22" stroke="#0284C7" stroke-width="2.5" stroke-linecap="round"/>
    <line x1="10" y1="50" x2="22" y2="50" stroke="#0284C7" stroke-width="2.5" stroke-linecap="round"/>
    <line x1="78" y1="50" x2="90" y2="50" stroke="#0284C7" stroke-width="2.5" stroke-linecap="round"/>
  </g>

  <!-- Typography: Brand Name -->
  <text x="135" y="54" font-family="-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif" font-size="34" font-weight="900" letter-spacing="-0.5" fill="#0F172A">
    METAL PROD
  </text>
  <text x="365" y="54" font-family="-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif" font-size="34" font-weight="300" letter-spacing="1" fill="#0284C7">
    SERVICE
  </text>

  <!-- Subtitle / Tagline -->
  <text x="137" y="84" font-family="-apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif" font-size="12.5" font-weight="600" letter-spacing="3" fill="#64748B">
    PRECISION CNC MACHINING • HEAT TREATMENT • SERIES • PROTOTYPES
  </text>
</svg>
```

3. Information Architecture & Navigation

The navigation bar is engineered for zero cognitive friction, intuitive 1-click accessibility, and high conversion.

3.1 Global Sticky Header Elements

```
[ BRAND LOGO ] ----- [ NAV ITEMS ] ---
[ LANGUAGE ] --- [ CTA BUTTON ]
Metal Prod Service   Home | About | Services | Catalogue | Equipment | Contact   [ EN | v ]
[ Request Offer ]
```

- 1. **Home:** One-click instant return to main screen / top scroll reset.
- 2. **About:** Deep dive into history, family heritage, engineering philosophy, and values.
- 3. **Services:** Dropdown and mega-menu encompassing: - CNC Milling (3-Axis & 5-Axis) - CNC Turning & Automatic Screw Machining (Strungire & Așchiere Automată) - Rapid Prototyping & Custom Unit Production - High-Volume Batch & Series Production - Technical Plastics Machining (POM, PTFE, PVC, PA6) - Industrial Quenching & Tempering Furnaces (Tratamente Termice)
- 4. **Catalogue:** Realized projects portfolio (custom shafts, flanges, screws, technical plastic parts, assemblies).
- 5. **Equipment:** Complete technical inventory of machine park, travel axes, furnace temperatures, and metrology labs.
- 6. **Contact:** Direct phone numbers, verified company emails, factory address, interactive map, working hours.
- 7. **Language Switcher:** Global dropdown with flags: **EN (Default), RO, HU, DE, FR, ES, IT.**
- 8. **Primary CTA:** Standout button `Request Offer / Cere Ofertă` anchored to fast multi-part RFQ configurator.

4. Multi-Language Architecture (i18n)

The application utilizes an instant ISO-639-1 routing and tokenized translation structure: - **Default Locale:** `en` (English) - **Supported Locales:** `ro` (Romanian), `hu` (Hungarian), `de` (German), `fr` (French), `es` (Spanish), `it` (Italian)

Key Header & UI Translation Dictionary

UI Key	English (Default)	Romanian (RO)	German (DE)	Hungarian (HU)	French (FR)	Spanish (ES)	Italian (IT)
<code>nav.home</code>	Home	Acasă	Startseite	Főoldal	Accueil	Inicio	Home
<code>nav.about</code>	About Us	Despre Noi	Über Uns	Rólunk	À Propos	Sobre Nosotros	Chi Siamo
<code>nav.services</code>	Services	Servicii	Leistungen	Szolgáltatások	Services	Servicios	Servizi
<code>nav.catalogue</code>	Catalogue	Catalog Piese	Katalog	Katalógus	Catalogue	Catálogo	Catalogo
<code>nav.equipment</code>	Equipment	Echipamente	Maschinenpark	Géppark	Équipements	Equipamiento	Macchinari
<code>nav.contact</code>	Contact	Contact	Kontakt	Kapcsolat	Contact	Contacto	Contatti
<code>cta.quote</code>	Request Offer	Cere Ofertă	Angebot anfragen	Ajánlatkérés	Demander un devis	Solicitar oferta	Richiedi preventivo

5. Complete Web Page Content & Copywriting

5.1 Home / Landing Page

Hero Section

- **Badge:** Precision Engineered in the EU • Certified ISO 9001:2015 Standards
- **Main Heading:** High-Precision CNC Machining, Custom Prototyping & Series Production
- **Sub-heading:** Built on family dedication, engineering seriousness, and technological mastery. We transform solid metals and technical plastics into micron-accurate components with certified in-house heat treatment.
- **Action Buttons:**
 - [Primary CTA] **Request a Free Technical Quote (CAD/Technical Drawing Analysis in 24h)**
 - [Secondary CTA] **Explore Our Plant & Machine Park**
- **Trust Badges:**
 - ± 0.005 mm Machining Tolerance
 - 100% In-House Heat Treatment & Quenching
 - 1 to 100,000+ Prototypes to High-Volume Series
 - 48-Hour Fast Track Rapid Prototyping Dispatch

Visual Assets Integration:

- **Hero Image:** `FactoryPhoto.JPG` (Showcases our external industrial facility, clean yard, modern infrastructure, and logistical access).
- **Secondary Feature Showcase:** `Factory_Interior.JPG` (Showcases our organized production floor, active CNC milling centers, clean oil-mist filtration systems, and operator consoles).

5.2 About Us (Tradition, Heritage & Values)

The Story of Metal Prod Service

"True precision is not just about programmable machines; it is about the dedication, character, and honesty of the people who operate them."

Metal Prod Service was established with a foundational mission: to provide the European and domestic mechanical industries with uncompromising precision parts manufactured with familial care, punctuality, and competitive pricing.

Over decades of generational industrial experience, we expanded from basic mechanical workshops into a high-technology CNC machining and thermal treatment facility. As an independent, family-owned business, we do not view clients as simple order numbers. Every project—from a single custom prototype shaft for a broken production line to 50,000 serialized precision screws—receives personal oversight from our senior engineering team.

Why Clients Rely on Our Family Enterprise:

1. **Uncompromising Seriousness & Integrity:** When we quote a delivery date, we meet it. When we guarantee a $H7$ tolerance or a 60 HRC surface hardness, every unit is certified.
2. **Direct Engineer-to-Engineer Dialogue:** No layered corporate bureaucracy. Our clients speak directly with tooling and machining specialists who understand CAD models, thermal expansion, and mechanical stresses.
3. **Fair Market Pricing:** Family-owned financial agility means zero inflated overheads. We pass operational efficiencies directly to our clients.
4. **Dedication to Perfection:** We inspect the first part, the middle part, and the final run with equal rigor using calibrated micrometric gauges and CMM arms.

5.3 Services & Manufacturing Capabilities

(Engineered with direct reference to the technical standards of piese-cnc.ro expanded with full manufacturing details)

Service 1: Precision CNC Milling (3-Axis & 5-Axis)

- **Overview:** High-speed, ultra-rigid milling of complex geometries, prismatic, deep pockets, and contouring.
- **Materials Processed:**
 - Structural & Carbon Steels: OL37, OL52, C45, 42CrMo4.
 - Stainless Steels: AISI 304, AISI 316L, AISI 303 (free machining), AISI 420.
 - Aluminum Alloys: Al 6060, 6082-T6, Al 7075 (Ergal / Aircraft Grade).
 - Copper & Brass: Cu-ETP, CW614N (Ms58).
- **Tolerances:** Standard ± 0.01 mm; Ultra-precision down to ± 0.005 mm upon specification.
- **Surface Finish:** Roughness values down to $Ra\ 0.8 - 0.4\ \mu m$ directly off the spindle.

Service 2: CNC Turning & Automatic Mechanical Machining (Strungire & Așchiere Automată)

- **Overview:** High-throughput cylindrical turning, grooving, boring, threading, and part-off operations.
- **Capabilities:**
 - Bar feeder continuous production for serial turned parts (bushings, pins, custom screws, hydraulic connectors).
 - Live tooling for combined turn-mill operations without secondary fixture setups.
 - Diameters from $\varnothing\ 3$ mm up to $\varnothing\ 450$ mm.

Service 3: Rapid Prototyping & Custom 1-Off Machining (Piese Unicat & Prototipuri)

- **Overview:** Rapid iteration service for R&D engineers, mechanical designers, and repair operations.
- **Turnaround:** Delivery in as fast as 48 to 72 hours following 3D CAD/STEP validation.
- **DFM (Design for Manufacturability):** Comprehensive review of radius sizes, wall thicknesses, and tool access to reduce unit production costs before cutting metal.

Service 4: High-Volume Serial Production (Producție CNC în Serie)

- **Overview:** Industrialized manufacturing for orders from 500 to 100,000+ pieces with dedicated fixturing, multi-clamping vices, and statistical process control (SPC).
- **Key Products:** Specialized fasteners, bolts, custom screws, bushings, spacers, manifolds, automotive brackets.

Service 5: Technical Plastics CNC Machining (Prelucrare Plastic Tehnic CNC)

(Detailed technical specification model inspired by piese-cnc.ro)

Engineering plastics are rapidly replacing metals across robotics, food handling, medical devices, and chemical processing due to weight reduction, chemical inertness, and zero corrosion. We machine:

1. **POM (Polyoxymethylene / Delrin / Acetal):** - *Properties:* High mechanical strength, exceptional rigidity, superior dimensional stability, low moisture absorption, excellent friction coefficient. - *Applications:* Precision gears, wear strips, bearing cages, electrical insulators, fluid valves.
2. **PTFE (Polytetrafluoroethylene / Teflon):** - *Properties:* Extreme thermal operating envelope ($-200^{\circ}C$ to $+260^{\circ}C$), near-universal chemical inertness, lowest coefficient of friction. - *Applications:* Cryogenic seals, high-temperature chemical gaskets, laboratory valve bodies, dielectric insulators.
3. **PVC (Polyvinyl Chloride - Industrial):** - *Properties:* Cost-effective, rigid, self-extinguishing, excellent resistance to concentrated acids, bases, and salts. - *Applications:* Chemical storage pump parts, galvanic bath fittings, drainage manifolds, industrial enclosures.

4. **PA6 / PA6-G (Polyamide 6 / Nylon):** - *Properties:* Extreme shock absorption, vibration damping, high wear resistance under abrasive friction. - *Applications:* Heavy-load rollers, guide wheels, conveyor sprockets, impact buffers, industrial pulleys.

Service 6: Professional Metal Heat Treatment & Quenching (Cuptoare Profesionale de Călire)

Unlike many machine shops that rely on third-party heat treaters (causing delays and dimensional warping), **Metal Prod Service** operates **in-house controlled industrial furnaces:** - **Processes:** - **Quenching (Călire):** Heating parts to austenitizing temperatures (800°C – 1050°C) with controlled soak times, followed by rapid quenching in oil, polymer, or forced atmosphere to lock in maximum hardness. - **Tempering (Revenire):** Secondary heating cycles (180°C – 650°C) to relieve internal stresses, eliminate brittleness, and achieve the exact Rockwell hardness (HRC) required by technical blueprints. - **Case Hardening / Carburizing (Cementare):** Creating ultra-wear-resistant surface layers (58–62 HRC) while preserving a ductile, shock-absorbing core. - **Stress Relieving (Detensionare):** Thermal stabilization before and after heavy machining to eliminate mechanical distortion in long shafts and thin plates. - **Furnace Atmosphere:** Electronically monitored thermal controllers with multi-point thermocouple logging ensuring uniform heating curves without oxidation or surface decarburization.

5.4 Machining Equipment & Plant Technology

Our investment into production hardware ensures maximum rigidity, accuracy, and zero downtime:

Equipment Category	Machine Model / Type	Technical Parameters & Working Envelope	Primary Capabilities
CNC Vertical Milling	Heavy 3-Axis & 4-Axis Centers (e.g., Cormak Mill 530 Class & High-Torque Spindles)	X: 800 mm, Y: 500 mm, Z: 550 mm; Spindle speeds up to 12,000 RPM; Rapid feeds 30 m/min	Prismatic parts, tight-tolerance tooling, complex mold plates, rapid drilling & tapping
CNC Turning Centers	High-Precision Slant Bed Lathes with Live Tooling & Automatic Bar Feeder	Max turning diameter: Ø 380 mm; Turning length: 650 mm; Bar feeder passage: Ø 65 mm	Fast mass-production of turned pins, bushings, threaded custom shafts, hydraulic fittings
Industrial Heat Treatment Furnaces	High-Temperature Chamber Furnaces with Microprocessor PID Profiling	Operating temp: Up to 1200°C; Chamber volume: 200 L; Multi-stage digital temperature ramping	Hardening, tempering, annealing, and stress relief for carbon steels, tool steels (1.2379, 42CrMo4, etc.)
Bandsawing & Blank Prep	Automatic Heavy-Duty Mitering Bandsaws	Cutting capacity up to Ø 320 mm; Automatic bar advance	High-speed, clean-cut preparation of raw billet materials
Metrology & Quality Assurance	Mitutoyo Digimatic Micrometers, Bore Gauges, Granite Surface Tables, Optical Hardness Testers	Resolution to 0.001 mm; Portable Rockwell / Brinell hardness testing calibrated to ISO 6508	100% Dimensional inspection, statistical validation, surface roughness measurement, hardness certificates

5.5 Product Catalogue & Portfolio Showcase

The catalogue represents precision parts manufactured for our aerospace, automotive, industrial machinery, and medical clients.


```

+-----+
| FILTER: [ All ] [ Custom Screws & Fasteners ] [ Shafts & Bushings ] [ Plastics ] |
+-----+
| [ IMG: Calibrated Metric & Trapezoidal Screws ] |
| Part: Custom High-Tensile Ground Lead Screws |
| Material: 42CrMo4 Steel + Quenched & Tempered (32-36 HRC) |
| Application: Heavy CNC Axis Drive Assemblies • Batch Size: 2,500 pcs |
+-----+
| [ IMG: Technical POM-C Gears & Bushings ] |
| Part: High-Speed Silent Conveyor Pinion |
| Material: Virgin POM (Polyoxymethylene Delrin) |
| Application: Food Processing Automation • Tolerances: ISO js6 |
+-----+
| [ IMG: Hardened Precision Splined Drive Shaft ] |
| Part: Splined Transmission Shaft |
| Material: 16MnCr5 Steel • Case Hardened (Cementat) 60±2 HRC |
| Application: Agricultural & Industrial Gearboxes • Total Runout: < 0.008 mm |
+-----+
| [ IMG: Industrial PTFE Sealing Flange ] |
| Part: High-Temperature Cryogenic Seal Flange |
| Material: Pure PTFE (Teflon) |
| Application: Chemical Valve Body Assembly • Operating Temp: -180°C to +240°C |
+-----+
| [ IMG: CNC Milled Aerospace Sensor Housing ] |
| Part: Ultra-Lightweight Drone Gimbal Bracket |
| Material: Aluminum 7075-T6 (Ergal) + Black Anodized |
| Application: Aerial Robotics & Defense Inspection • Weight: 42 grams |
+-----+
| [ IMG: Large Stainless Steel Manifold Block ] |
| Part: High-Pressure Hydraulic Distribution Block |
| Material: Stainless Steel AISI 316L (Acid Resistant) |
| Application: Marine & Offshore Hydraulic Systems • Pressure Tested: 350 Bar |
+-----+

```

5.6 Request a Quote (RFQ Engine)

An interactive, high-converting 3-step quotation portal: 1. **Part Specifications:** - Production Mode: ☐ Single Prototype / Unicat ☐ | ☐ Small Series (10-100 pcs) ☐ | ☐ High Volume Series (500-50,000+ pcs) ☐ - Material Classification: Steel / Stainless / Aluminum / Brass / POM / PTFE / PA6 / PVC / Other - Heat Treatment Needed: ☐ None ☐ | ☐ Quenching & Tempering (Călire) ☐ | ☐ Case Hardening (Cementare) ☐ | ☐ Stress Relief ☐ 2. **CAD & Technical Drawing Upload:** - Supported File Types: .STEP .STP .IGES .DWG .DXF .PDF .ZIP (Max 50MB per file). - Drag-and-drop target zone with encrypted transmission badge. 3. **Delivery & Contact Data:** - Company Name, Tax ID / CUI (for EU businesses), Engineer Contact, Email, Phone, Required Delivery Date. - Guaranteed response timeframe: **Engineering review & formal quote delivered within 24 business hours.**

5.7 Contact & Factory Location

- **Company Name:** Metal Prod Service S.R.L.
- **Factory Address:** Str. Industriei / Precision Industrial Park (Logistics access for 40-ton articulated trucks)
- **Direct Phone:** +40 (0) 7xx xxx xxx / Office: +40 (0) 2xx xxx xxx
- **Technical & RFQ Email:** rfq@metalprodservice.com / contact@metalprodservice.com
- **Operating Hours:** Monday – Friday: 07:00 – 19:00 (Production Floor in 2 shifts) | Saturday: 08:00 – 14:00 (Maintenance & Scheduled Dispatch)

6. Design System & 2026 UI/UX Standards

Adhering to modern manufacturing web design trends (interactive industrial showcases, clean typography, dark/light contrast modes, and zero bloat):

6.1 Color Architecture

```
:root {  
  /* Brand Industrial Colors */  
  --color-brand-primary: #0284C7;      /* Electric CNC Precision Blue */  
  --color-brand-primary-hover: #0369A1; /* Deep Milling Blue */  
  --color-brand-accent: #F59E0B;        /* Furnace Flame Amber (Heat Treatment) */  
  --color-surface-dark: #0F172A;        /* Slate / Machine Bed Anthracite */  
  --color-surface-card: #1E293B;        /* Secondary Industrial Dark */  
  --color-surface-light: #F8F9FC;       /* Clean Metrology White */  
  --color-text-main: #0F172A;           /* High Contrast Body Text */  
  --color-text-muted: #64748B;          /* Technical Specification Label Muted */  
  --color-border-subtle: #E2E8F0;        /* Precision Grid Border */  
  --color-border-dark: #334155;         /* Machine Bay Border */  
  --color-success: #10B981;             /* Tolerance Approved Green */  
}
```

6.2 Typography & Structural Hierarchy

- **Primary Typeface:** Inter, -apple-system, BlinkMacSystemFont, Segoe UI, Roboto (Crisp legibility for dimensional tables and technical data).
- **Monospace Technical Data Typeface:** JetBrains Mono, SF Mono, Menlo (Used for part numbers, ISO tolerances, Ra roughness values, and temperature specs).
- **Layout Rhythm:** Fluid 12-column grid with dynamic rem spacing based on 8pt design intervals.

7. Cross-Platform, Responsive & Foldable Layout Specs

The web platform is designed and tested for universal responsiveness across all modern devices and operating systems:

7.1 Breakpoints & Adaptive Matrix

1. **Foldable Devices & Dual-Screens (Samsung Galaxy Z Fold series, Google Pixel Fold, Microsoft Surface Duo):** - CSS Media Query: @media (spanning: single-fold-vertical), @media (spanning: single-fold-horizontal) - Dual-pane layout: Left pane displays technical parameters/furnace specs; right pane displays 3D CAD viewer / part renders.
2. **Compact Mobile (320px – 480px) (iPhone SE, iPhone 13/14/15/16 Mini, standard Androids):** - Header switches to sticky bottom thumb-friendly navigation bar or clean slide-in drawer. - 1-click WhatsApp / Direct Phone dialing sticky button.
3. **Mainstream Smartphones (481px – 768px):** - 2-column catalogue grids; collapsible technical data accordions.
4. **Tablets & Laptops (769px – 1280px) (iPad Pro, Surface Pro, Chromebooks):** - Full header with condensed navigation; instant flyout language menu.
5. **Ultra-Wide Workstations (1440px – 2560px+ 4K Monitors):** - Centered container capped at 1380px with edge gradient ambient glow, preventing stretched text lines.

7.2 Operating System & Browser Compatibility

- **Browsers:** Google Chrome (v115+), Apple Safari (iOS & macOS 16+), Mozilla Firefox (v115+), Microsoft Edge (Chromium), Opera, Samsung Internet.

- **Operating Systems:** iOS, Android (v10–15+), Windows 10/11, macOS, Linux (Ubuntu, Fedora, Arch via Chromium/Firefox).

8. Comprehensive Testing & Quality Assurance Plan

To guarantee a defect-free, enterprise-ready web platform, a 5-tier QA testing regimen is defined:

TIER 5: SECURITY & COMPLIANCE
OWASP Top 10 • File Upload Protection • GDPR / Cookie Policy
TIER 4: REGRESSION & INTEGRATION
E2E Playwright Flows • RFQ Form Handling • Language Fallbacks
TIER 3: PERFORMANCE & LOAD STRESS
Lighthouse 98+ • Core Web Vitals • K6 1,000 Concurrent Users
TIER 2: UI / UX & CROSS-BROWSER
Foldable Resizing • Touch Targets • WCAG 2.1 AA Contrast
TIER 1: FUNCTIONAL UNIT TESTING
Buttons • Links • Forms • Dropdowns • Language Switching

8.1 Functional & Interactive Button Testing Matrix

Every interactive element has an explicit assertion and verification routine:

Component	Target Action	Expected Result	Pass Criteria
Logo (Header)	Click / Tap	Instant reset to Home <code>#top</code> without full-page re-fetch	URL anchors to <code>/</code> or <code>/#top</code> ; zero page jitter
Nav: "Home"	Click / Tap	Smooth scroll to Hero view	Viewport displays Hero within 200ms
Nav: "About"	Click / Tap	Scrolls to Family Heritage & Tradition section	Section <code>#about</code> centered in viewport
Nav: "Services"	Hover / Click	Reveals 6 sub-service cards with technical details	Sub-links accessible via keyboard navigation
Nav: "Catalogue"	Click / Tap	Opens part grid with functional material filters	Filter toggle redraws grid in < 50ms
Nav: "Equipment"	Click / Tap	Anchors to machine specifications & furnace table	Equipment table fully visible
Nav: "Contact"	Click / Tap	Anchors to factory contact details & map	Map iframe and direct phone links active
Language Select	Select RO/DE/HU/FR/ES/IT	Dynamic i18n swap of all headings, tables, labels	Zero untranslated strings; correct locale tag in HTML
RFQ "Submit"	Click without files	Trigger HTML5 & custom JS field validation	Red outline on mandatory fields with clear hints
RFQ File Dropper	Drop STEP/PDF file	Display file name, size, and green validation tick	Rejects invalid extensions (.exe, .bat, .sh)

8.2 Performance, Core Web Vitals & Load Testing

- **Lighthouse Performance Benchmarks (Year 2026 Target):**
- **Performance:** $\geq 98 / 100$
- **Accessibility:** 100 / 100 (WCAG 2.1 AA compliant)
- **Best Practices:** 100 / 100
- **SEO:** 100 / 100
- **Core Web Vitals Thresholds:**
- **LCP (Largest Contentful Paint):** < 1.2 seconds on 4G network.
- **FID / INP (Interaction to Next Paint):** < 50 milliseconds.
- **CLS (Cumulative Layout Shift):** 0.000 (Zero layout jump upon image load via explicit `aspect-ratio`).
- **Load & Concurrency Testing (k6 / Artillery Protocol):**
- **Scenario:** 1,200 concurrent virtual users browsing catalogue pages and submitting CAD RFQ inquiries simultaneously.
- **Threshold:** 0% packet loss, 99th percentile response time < 350 ms, zero database locks.

8.3 UI / UX & Cross-Browser Compatibility Testing

- **Viewport Testing:** Tested across 320px (iPhone SE), 375px (iPhone mini), 390px (iPhone 14/15/16), 768px (iPad Mini), 820px (iPad Air), 1024px (iPad Pro), 1920px (Full HD Desktop), 3840px (4K Ultrawide).
- **Foldable Testing:** Fold/unfold transition handling on Samsung Galaxy Fold displays with zero CSS clipping.
- **Dark / Light Contrast:** Tested using WebAIM contrast analyzer ensuring minimum 4.5:1 ratio for normal body copy and 3.0:1 for heavy technical headings.

8.4 Non-Functional, Scalability & Regression Testing

- **Automated Regression Suite:** Continuous Integration (CI) pipeline running Playwright end-to-end tests on every code commit.
- **Failover & Redundancy:** Static site generation (SSG) with CDN edge deployment (Cloudflare / AWS CloudFront) ensuring 99.99% uptime even during factory line power fluctuations.

8.5 Cybersecurity, Data Integrity & RFQ Upload Protection

Industrial CAD models are sensitive intellectual property. The RFQ architecture includes enterprise-grade safeguards: 1. **File Sanitization:** All incoming CAD/PDF files are validated by magic byte header checking to block disguised executable payloads. 2. **End-to-End Encryption:** HTTPS enforced with TLS 1.3, HSTS preloading, and AES-256 server-side encryption for client drawings. 3. **Form Bot Protection:** Cloudflare Turnstile (zero-friction captcha) integrated to eliminate automated form spam without requiring frustrating user captchas. 4. **GDPR / EU Data Protection Compliance:** Explicit consent checkboxes for technical data processing, cookie consent management, and instant data deletion request triggers.

9. Technical Implementation Stack & Deployment

Recommended 2026 Production Technology Stack

- **Framework:** Next.js 15 (App Router) or Vite + Vanilla TypeScript for high-performance static rendering.
- **Styling Engine:** Tailwind CSS v4 with custom CSS variables for industrial colors.
- **Icons:** Lucide Icons (clean, high-precision engineering glyphs).
- **Hosting / Edge CDN:** Vercel Edge Network or AWS S3 + CloudFront with global points of presence in Frankfurt, Bucharest, London, and New York.
- **Lead Integration:** Webhook delivery forwarding incoming RFQ specifications and CAD attachments directly to the workshop engineering inbox (`rfq@metalprodservice.com`) and enterprise ERP.

Image Asset Placeholder References:

- **FactoryPhoto.JPG** : Embedded in the Hero and Factory Facility section showing the modern industrial exterior building and logistical truck docks.
- **Factory_Interior.JPG** : Embedded in the Equipment and Production Floor section showing the precision CNC milling lines, automated lathe batteries, and in-house heat treatment furnace cells.

End of Architecture Specification.